

Paper 05 — Communication Protocol

WebSocket, OSC and the C++↔React Bridge

WHYCREMISI RESEARCH PAPERS – N.05
Internal Communication Protocol (JSON v1.1)

Message Structure

```
{ "type": "category.action", "id": "uuid", "timestamp": 0, "payload": {} }
```

UI → Plugin Messages

Type	Key Payload Fields
ai.prompt	prompt, provider
daw.command	command, trackId, valueDb, bpm
plugin.control ★	trackId, pluginSlot, pluginName, paramName, value
plugin.query ★	trackId, pluginSlot, paramName
config.set	key, value, provider
plugin.init	version, capabilities

Plugin → UI Messages

Type	Key Payload Fields	Frequency
ai.stream	chunk, isDone	Per word/group
ai.response	status, content, model, durationMs	Once per prompt
daw.transport	isPlaying, bpm, positionSeconds	On change + 1Hz
daw.meter	trackId, leftDb, rightDb, peakLeftDb	~30fps
plugin.state ★	pluginName, parameters[]	On query/change
agent.memory ★	event, note	On memory update
plugin.error	code, message, severity	On error

★ = New in v1.1

OSC Addresses (DAW Communication)

```
/transport/play|stop|record
/transport/bpm [float]
/track/[n]/volume|pan|mute|solo [value]
/fx/[n]/[m]/param/[p] [float]
/master/volume|gain [float]
```

Latency Targets

Message type	Target	Measured
WebSocket round-trip	< 5ms	2–3ms
AI TTFB (local Ollama)	< 1000ms	600–800ms
AI TTFB (cloud)	< 600ms	300–500ms
Meter update (30fps)	33ms	33ms ✓
Plugin param change	< 2ms	1ms ✓

→ Continue: [Paper 06 — Implementation Roadmap](#)